SQL Module End Assignment

E-Commerce Customer Churn Analysis

Dataset :

https://drive.google.com/uc?export=download&id=1iKKCze_Fpk2n_g3BIZBiSjcDFdFcEn3D

Data Cleaning:

Step1: Handling Missing Values

```
SELECT COUNT(*) FROM customer_churn WHERE HourSpendOnApp IS NULL;
SELECT COUNT(*) FROM customer_churn WHERE OrderAmountHikeFromlastYear IS NULL;
SELECT COUNT(*) FROM customer_churn WHERE DaySinceLastOrder IS NULL;
 SELECT COUNT(*) FROM customer_churn WHERE HourSpendOnApp IS NULL;
SELECT COUNT(*) FROM customer_churn WHERE OrderAmountHikeFromlastYear IS NULL;
SELECT COUNT(*) FROM customer_churn WHERE DaySinceLastOrder>100;
select * from customer_churn where WarehouseToHome >100;
SELECT COUNT(*)
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	COUNT(*)
▶	0

No missing values were found in the specified columns; hence, imputation like mean, mode was not required.

Step2: Outliers

```
SELECT COUNT(*)
FROM customer_churn
WHERE WarehouseToHome > 100;
SELECT DISTINCT PreferredPaymentMode FROM customer_churn;
UPDATE customer_churn
SET PreferredPaymentMode = 'Cash on Delivery'
WHERE PreferredPaymentMode = 'COD';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	COUNT(*)
▶	2

```
delete from customer_churn where  WarehouseToHome > 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	COUNT(*)
▶	0

2 outliers were identified in the WarehouseToHome column (values >100) and removed.

Dealing with inconsistences

Step1:

```
SELECT DISTINCT PreferredLoginDevice FROM customer_churn;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	PreferredLoginDevice
▶	Mobile Phone
	Computer

Step2:

```
5688 •  select distinct PreferedOrderCat from customer_churn;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PreferedOrderCat
Laptop & Accessory
Mobile Phone
Others
Fashion
Grocery

Refresh data re-executing the original query

Replaced occurrences of “Phone” in the 'PreferredLoginDevice' column and

“Mobile” in the 'PreferedOrderCat' column with “Mobile Phone” to ensure

uniformity.

Standardized payment mode values by Replacing "COD" with "Cash on Delivery" and

"CC" with "Credit Card" in the PreferredPaymentMode column.

Data Transformation

Step1: Column Renaming

➢ Rename the column "PreferedOrderCat" to "PreferredOrderCat".

➢ Rename the column "HourSpendOnApp" to "HoursSpentOnApp".

```
5690 •  alter table customer_churn rename column HourSpendOnApp to HoursSpendOnApp;
5691 •  alter table customer_churn rename column PreferedLoginDevice To PreferredLoginDevice;
5692
5693
```

Column renamed to the required name

Step2: Creating New Columns:

a) Create a new column named ‘ComplaintReceived’ with values "Yes" if the corresponding value in the ‘Complain’ is 1, and "No" otherwise.
b) Create a new column named 'ChurnStatus'. Set its value to “Churned” if the corresponding value in the 'Churn' column is 1, else assign “Active”.

```
update customer_churn set ComplaintReceived = case when Complain =1 then 'yes' else 'no ' end;
update customer_churn set ChurnStatus = case when Churn = 1 then 'Churned' else 'Active' end;
select ComplaintReceived, ChurnStatus from customer_churn;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

ComplaintReceived	ChurnStatus
yes	Churned
yes	Churned
yes	Churned
no	Churned
no	Churned
yes	Churned

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Created new columns and assigned the values using conditions

Column Dropping:

Drop the columns "Churn" and "Complain" from the table.

```
alter table customer_churn drop column Churn, drop column Complain;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode
50001	4	Mobile Phone	3	6	Debit Card
50002	NULL	Phone	1	8	UPI
50003	NULL	Phone	1	30	Debit Card
50004	0	Phone	3	15	Debit Card
50005	0	Phone	1	12	CC

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The columns ‘Churn’ and ‘Complain’ were successfully removed

Data Exploration and Analysis:

Step1: Retrieve the count of churned and active customers from the dataset.

```
select ChurnStatus, count(*) from customer_churn group by ChurnStatus ;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ChurnStatus	count(*)
Churned	948
Active	4682

Out of 5630 customers, 948 customers have churned while 4682 customers remain active, indicating a churn rate of approximately 16.8%

Step2: Display the average tenure and total cashback amount of customers who churned.

```
select sum(CashBackAmount) Total_Cashback, avg(Tenure) Avg_Tenure  from customer_churn where ChurnStatus='Churned';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Cashback	Avg_Tenure
152030	3.3795

Step3: Determine the percentage of churned customers who complained.

```
select count(*) from customer_churn where ComplaintReceived= 'Yes' and ChurnStatus = 'Churned' ;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

count(*)
508

```
select count(*) from customer_churn where ChurnStatus='Churned';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

count(*)
948

Complaint % = (508 / 948) * 100 = 53.6%

percentage of churned customers who complained =53.6%

Step4: Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessory.

```
select count(*) Highest_citytier from customer_churn where ChurnStatus = 'Churned'
and PreferedOrderCat = 'Laptop & Accessory' group by CityTier order by Highest_citytier Desc Limit 1;

select  PreferredPaymentMode, count(*) Highest_Paymentmode from customer_churn where ChurnStatus = 'Active
group by PreferredPaymentMode order by Highest_Paymentmode desc limit 1;
select sum(OrderAmountHikeFromlastYear) Total_orderamount from customer_churn
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Highest_citytier
150

city tier with the highest number of churned customers whose

preferred order category is Laptop & Accessory identified is 150

Step5: Identify the most preferred payment mode among active customers.

```
select  PreferredPaymentMode, count(*) Highest_Paymentmode from customer_churn where ChurnStatus = 'Active'
group by PreferredPaymentMode order by Highest_Paymentmode desc limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

PreferredPaymentMode	Highest_Paymentmode
Debit Card	1958

Most preferred Payment mode among active customers is found Debitcard

Step6: Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.

```
select sum(OrderAmountHikeFromlastYear) Total_orderamount from customer_churn
where MaritalStatus = 'Single' and PreferredLoginDevice = 'Mobile Phone';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_orderamount
18953

The total order amount hike from last year for customers who are single and prefer ordering from mobile phones is 18953

Step7: Find the average number of devices registered among customers who used UPI as their preferred payment mode.

```
select avg(NumberOfDeviceRegistered) Average_Devices from customer_churn where PreferredPaymentMode = 'UPI';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Average_Devices
3.7174

The average number of devices registered among customers who used UPI as their preferred payment mode is 3.7174

Step8: Find the average number of devices registered among customers who used UPI as their preferred payment mode.

```
5711 • select CityTier, count(*) Highestno_customers from customer_churn group by CityTier order by Highestno_customers = ' Active' desc limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

CityTier	Highestno_customers
3	1722

City Tier 3 has the highest number of customers, with a total of 1,722 customer

Step9: Identify the gender that utilized the highest number of coupons.

```
5713 • select sum(Couponused) Highestno_couponused, Gender from customer_churn group by Gender order by Highestno_couponused desc Limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Highestno_couponused	Gender
5497	Male

Gender Category that utilized the highest number of coupons IS Male 5497

Step10: List the number of customers and the maximum hours spent on the app in each preferred order category.

```
5714 • select PreferredOrderCat, count(*) Total_custspent , max(HoursSpendOnApp) Maxhrsspent from customer_churn group by PreferredOrderCat;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PreferredOrderCat	Total_custspent	Maxhrsspent
Laptop & Accessory	2050	5
Mobile	809	4
Mobile Phone	1271	5
Others	264	4
Fashion	826	5
Grocery	410	4

Step11: Calculate the total order count for customers who prefer using credit cards and

have the maximum satisfaction score.

```
5715 • select count(*) OrderCount from customer_churn where PreferredPaymentMode = 'Credit Card'
5716   and SatisfactionScore = (select max(SatisfactionScore) from customer_churn);
5717
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

OrderCount
364

Total Order count using credit cards with maximum statisfaction score is 364

Step12: What is the average satisfaction score of customers who have complained?

```
select avg(SatisfactionScore) from customer_churn where ComplaintReceived = 'yes';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ComplaintReceived
NULL

No customers have recorded complaints, so the average satisfaction score cannot be calculated

Step13: List the preferred order category among customers who used more than 5 coupons.

```
select avg(SatisfactionScore) from customer_churn where Complain = 1;
select PreferedOrderCat from customer_churn where CouponUsed>5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

avg(SatisfactionScore)
2.9988

The average satisfaction score of customers who raised complaints is approximately 3.0, indicating a moderate level of satisfaction.

Step14: List the preferred order category among customers who used more than 5 coupons.

```
select distinct PreferedOrderCat from customer_churn where CouponUsed>5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PreferedOrderCat
Others
Laptop & Accessory
Fashion
Mobile Phone
Grocery

Step15: List the top 3 preferred order categories with the highest average cashback amount.

```
select PreferedOrderCat, avg(CashbackAmount) highest_avgamount from customer_churn
group by PreferedOrderCat order by highest_avgamount desc limit 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

PreferedOrderCat	highest_avgamount
Others	304.4545
Grocery	266.2366
Fashion	210.4031

The top 3 preferred order categories with the highest average cashback amount was listed

Step16: Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.

```
select PreferredPaymentMode, avg(Tenure) as Tenureavg_10m from customer_churn where OrderCount >500
group by PreferredPaymentMode having (Tenureavg_10m) =10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PreferredPaymentMode	Tenureavg_10m

No preferred payment modes having orders greater than 500

Step17: Categorize customers based on their distance from the warehouse to home such as 'Very Close Distance' for distances <=5km, 'Close Distance' for <=10km, 'Moderate Distance' for <=15km, and 'Far Distance' for >15km. Then, display the churn status breakdown for each distance category.

```
select case when WarehouseToHome <=5 then 'very close distance'
when WarehouseToHome <=10 then 'close distance' when WarehouseToHome <=15 then 'moderate distance'
else 'far distance' end as distancecategory , ChurnStatus, count(*) Total_customers
 from customer_churn group by distancecategory, ChurnStatus;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

distancecategory	ChurnStatus	Total_customers
far distance	Churned	498
moderate distance	Churned	184
close distance	Active	1696
moderate distance	Active	1106
far distance	Active	1873
very close distance	Active	7
very close distance	Churned	1

Step18: List the customer's order details who are married, live in City Tier-1, and their order counts are more than the average number of orders placed by all customers.

```
select MaritalStatus, CityTier, OrderCount from customer_churn where MaritalStatus = 'Married' and CityTier = 1 and
OrderCount > (select avg(OrderCount) from customer_churn);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MaritalStatus	CityTier	OrderCount
Married	1	6
Married	1	4
Married	1	7
Married	1	6
Married	1	9
Married	1	6
Married	1	12

Step19: a) Create a 'customer_returns' table in the 'ecomm' database and insert the following data:

ReturnID CustomerID ReturnDate RefundAmount

1001 50022 2023-01-01 2130

1002 50316 2023-01-23 2000

1003 51099 2023-02-14 2290

1004 52321 2023-03-08 2510

1005 52928 2023-03-20 3000

1006 53749 2023-04-17 1740

1007 54206 2023-04-21 3250

1008 54838 2023-04-30 1990

```
create table customer_returns ( ReturnID int primary key, CustomerID int, ReturnDate varchar(50), RefundAmount int );
insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) values(1001, 50022, '2023-01-01', 2130);
insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) values(1002, 50316, '2023-01-23', 2000);
insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) values(1003, 51099, '2023-02-14', 2290);
 insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) values(1004, 52321, '2023-03-08', 2510);
 insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) values(1005, 52928, '2023-03-20', 3000);
 insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) values(1006, 53749, '2023-04-17', 1740);
 insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) values(1007, 54206, '2023-04-21', 3250);
 insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) values(1008, 54838, '2023-04-30', 1990);
```

Output

Action Output

#	Time	Action	Message
9	19:18:43	insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) valu...	1 row(s) affected
10	19:18:43	insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) valu...	1 row(s) affected
11	19:18:43	insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) valu...	1 row(s) affected
12	19:18:43	insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) valu...	1 row(s) affected
13	19:18:43	insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) valu...	1 row(s) affected
14	19:18:43	insert into customer_returns(ReturnID, CustomerID, ReturnDate, RefundAmount) valu...	1 row(s) affected

Created a customer_returns table and inserted the given data

b) Display the return details along with the customer details of those who have churned and have made complaints.

```
select ReturnID, ReturnDate, RefundAmount, Complain, ChurnStatus
from customer_returns s
 inner join customer_churn c
on c.CustomerID=s.CustomerID
where c.ChurnStatus = 'Churned' and c.Complain=1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ReturnID	ReturnDate	RefundAmount	Complain	ChurnStatus
1002	2023-01-23	2000	1	Churned
1004	2023-03-08	2510	1	Churned
1006	2023-04-17	1740	1	Churned